

# SHYAM PATHAK

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## SUMMARY

AI Engineer focused on Generative AI and real-world ML systems, with hands-on experience building production-grade RAG pipelines, LLM-powered agents, and computer vision solutions. Developed end-to-end AI systems for legal document intelligence and engineering data extraction using FastAPI, LangChain, and vector databases. Strong in designing scalable pipelines from data processing to deployment, with emphasis on accuracy, efficiency, and reducing hallucination in LLM outputs.

## EXPERIENCE

**Junior AI Engineer Intern, Masin AI** 09/2025 – Present | Gurugram

- Built end-to-end RAG-based pipelines for legal document analysis using LLMs, embeddings, and vector databases, achieving 92% retrieval accuracy and reducing hallucinations through citation grounding
- Developed LLM-powered conversational agents using LangChain and LangGraph, enabling context-aware multi-step reasoning and improving response relevance and system efficiency
- Engineered a computer vision pipeline leveraging YOLO and OCR (Tesseract/PaddleOCR) to detect, extract, and quantify components from engineering drawings
- Designed scalable document processing architecture (chunking, embedding generation, semantic retrieval) for efficient handling of large documents with low-latency querying
- Contributed to modular AI system deployment using FastAPI and async processing, improving performance, maintainability, and scalability across pipeline components

## PROJECTS

**Pipe Takeoff AI** , AI Engineer Project 11/2025 – 03/2025

- Built an end-to-end AI system to extract pipe quantities, engineering labels, and topology from PDF drawings using YOLO, OCR, and FastAPI
- Designed a multi-stage pipeline (PDF parsing → detection → OCR → scale normalization → quantity aggregation) for real-world engineering documents
- Implemented dual processing modes (vector-based fast mode <10s/page and OCR-based deep scan) to balance speed and accuracy
- Developed API-driven architecture with FastAPI and integrated interactive querying via RAG-based chat interface
- Achieved scalable, production-ready system design with modular components, testing pipelines, and deployment support (Docker)

**DataForge\_pro** , AI-Powered Data Platform 03/2026 – 04/2026

- Built a production-grade AI-powered data platform enabling users to clean, transform, query, and visualize datasets using natural language, leveraging FastAPI, Pandas, and DuckDB
- Designed an end-to-end data pipeline (ingestion → profiling → cleaning → transformation → SQL querying → visualization) supporting CSV/Excel inputs and real-time processing
- Implemented 20+ automated data cleaning operations, including IQR/Z-score outlier detection, intelligent null handling, schema inference, and data quality profiling
- Integrated LLM-powered intelligence (Google Gemini) for dataset understanding, natural language to SQL conversion, and generation of business insights with role-specific recommendations
- Developed an interactive analytics layer with dynamic visualizations and AI-driven chart interpretation, improving accessibility for non-technical users

**DrawVision-AI** , Computer Vision + LLM System 02/2026 – 03/2026

- Built an AI-powered system to extract structured piping specifications from engineering drawings using YOLO (object detection) and Gemini LLM for intelligent parsing
- Designed an end-to-end processing pipeline (PDF ingestion → image preprocessing → object detection → LLM-based information extraction → structured CSV/Excel output) for automated engineering data extraction
- Implemented real-time processing using WebSockets with live visual feedback, including bounding box annotations for detected components
- Developed a hybrid CV + LLM architecture to interpret complex engineering labels (diameter, service codes, line numbers), achieving ~90%+ extraction accuracy
- Enabled scalable and user-friendly interaction through a FastAPI backend and interactive frontend for visualization and export of results

## TECHNICAL SKILLS

|                                                                                               |                                                                                                                             |                                                                                                                       |                                                                                                         |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Programming &amp; Core</b><br>Python, SQL                                                  | <b>Databases &amp; Data Systems</b><br>FAISS, ChromaDB, SQLite, DuckDB, PostgreSQL (basic)                                  | <b>Backend &amp; APIs</b><br>FastAPI, REST APIs, WebSockets, Async Programming                                        | <b>Computer Vision</b><br>YOLO (v8/v26), OpenCV, OCR (Tesseract, PaddleOCR), Image Processing Pipelines |
| <b>Machine Learning</b><br>Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation | <b>Generative AI &amp; NLP</b><br>LLMS, RAG Pipelines, LangChain, Prompt Engineering, Embeddings, Hugging Face Transformers | <b>Deployment &amp; DevOps</b><br>Docker, Git, GitHub, Railway/Render, Environment Management, CI/CD (basic), VS Code |                                                                                                         |

## EDUCATION

**Jaypee University of Engineering and Technology, B.Tech in Computer Science** 10/2021 – 05/2025 | Guna

**KR Mangalam World School, Class XII (CBSE) – 82%** 2020 – 2021 | Gurugram

**City Montessori School, Class X (ICSE) – 92%** 2014 – 2019 | Lucknow

## LEADERSHIP & CERTIFICATE

**Bitwise Club, Secretary**

- Participated in coding challenges, mentored 50+ students

**Google Students Developers Club, Member**

- Practiced new skills, coding challenges, and organized tech events

[link to all my certificates](#)